

# RAJESH NAUTIYAL

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## PROFESSIONAL SUMMARY

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Network Automation and AI-Enabled Validation Lead with 14+ years of experience driving network automation, AI-assisted troubleshooting, regression validation, and infrastructure operations for Cisco enterprise networking platforms. Proven expertise in Python, PyATS, CI/CD, Cisco NXOS/IOS-XE, and large-scale automation ecosystems supporting 150+ devices and 500+ automation scripts. Experienced in applying AI technologies, domain-specific RAG systems, Copilot, Codex, and agent-assisted workflows to accelerate root cause analysis, validation, infrastructure management, and engineering productivity.

## CORE COMPETENCIES

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Network Automation | AI-Assisted Troubleshooting | Root Cause Analysis | Regression Validation | Network Programmability | PyATS | Cisco NXOS | Cisco IOS-XE | Infrastructure Automation | Domain-Specific RAG | Agent-Assisted Workflows | CI/CD Automation | Python Development | Test Infrastructure Management | Release Validation | Team Leadership

## TECHNICAL SKILLS

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**AI & Agentic Automation:** GitHub Copilot, Codex, Prompt Engineering, Agent-Assisted Workflow Design, Domain-Specific RAG, AI-Assisted Development, AI-Based Root Cause Analysis, Conversational Knowledge Systems, AI-Driven Validation and Troubleshooting, LLM-Assisted Engineering Workflows

**Automation & Development:** Python, PyATS, Pytest, Django, TCL, Bash, Shell Scripting, YAML, JSON

**Network Platforms:** Cisco NXOS, IOS-XE, N7K, N9K, N9Kv, ASR1K, CAT9K, CSR1000v

**Network Programmability:** YANG Models, REST APIs, Model-Driven Programmability Validation, Device Configuration Validation, Telemetry Validation, API Response Validation

**Infrastructure & Cloud:** Linux RHEL7/RHEL8, Cisco UCS, VMware ESXi, AWS

**CI/CD & Tools:** Jenkins, Git, CVS, Visual Studio Code, CI/CD job execution and management

**Protocols:** VXLAN, BGP, OSPF, EIGRP, HSRP, VRRP, VLAN, STP, EVPN, MPLS

## PROFESSIONAL EXPERIENCE

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**CAPGEMINI** | Dec 2022 - Present

*Senior Professional II | Regression and Automation Lead | Project: Cisco*

- Lead regression and automation validation for Cisco NXOS and IOS-XE platforms, collaborating directly with Cisco engineering teams on enterprise networking releases.
- Lead a team of 5 engineers responsible for regression execution, automation maintenance, failure triage, release validation, and quality sign-off activities.
- Manage large-scale automation infrastructure comprising 150+ physical and virtual devices and 500+ automation scripts supporting multiple parallel release cycles.
- Own automation lab readiness, topology creation, hardware allocation, Linux administration, and Python-based validation environments.
- Designed and implemented an AI-powered regression support assistant enabling engineers to access regression insights, troubleshooting guidance, and operational knowledge on demand.
- Developed AI-assisted root cause analysis and failure triage solutions using domain-specific RAG (MdRAG), reducing investigation time from 1-2 hours to approximately 15 minutes.
- Built AI-assisted topology validation and infrastructure automation solutions for connectivity verification, device cleanup, baseline provisioning, and environment readiness.
- Leveraged GitHub Copilot, Codex, and AI-assisted development workflows to accelerate automation development, code remediation, debugging, and engineering productivity.
- Developed a Django-based infrastructure management application for Cisco UCS environments supporting virtual device lifecycle management and operational monitoring.
- Trained and mentored 10+ engineers on automation frameworks, AI-assisted troubleshooting, Cisco platform debugging, and operational best practices.

**TATA CONSULTANCY SERVICES (TCS)** | Mar 2012 - Dec 2022

*Associate Consultant | Technical Lead | Mar 2018 - Dec 2022 | Project: Cisco*

- Led automation and regression testing initiatives for Cisco NXOS and IOS-XE platforms.
- Tested, validated, and supported Python-based automation scripts using PyATS, YANG models, REST APIs, JSON/YAML parsing, and Cisco internal automation libraries.
- Set up environments and automation flows for model-driven programmability validation, including device configuration validation, telemetry validation, and API response validation.
- Managed automation labs, Linux servers, regression infrastructure, and Cisco UCS-based test environments.
- Supported regression testing and automation validation for protocol scripts covering VXLAN, BGP, OSPF, EIGRP, HSRP, VRRP, VLAN, STP, EVPN, and MPLS.
- Performed failure analysis, root cause support, defect tracking, and quality validation for Cisco platform releases.
- Coordinated automation integrations and release validation activities across multiple Cisco platform teams.

## **TATA CONSULTANCY SERVICES (TCS) | May 2012 - Feb 2018**

*Automation Testing Engineer | Project: Cisco*

- Developed TCL automation scripts for router and switch command validation.
- Performed regression testing and debugging support for Cisco NXOS and IOS-XE platforms.
- Worked on L2/L3 protocol script regression testing and automation validation.
- Analyzed device logs, test failures, automation errors, and regression results to identify issues.
- Supported defect reporting and issue tracking using Cisco internal defect management tools.

## **SELECTED PROJECTS**

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**AI-Powered Network Validation and Failure Triage Solution:** Developed an AI-assisted triage and RCA solution to analyze Jenkins failures, traceback errors, device logs, regression result files, and historical defect data. The solution identifies probable root causes, suggests next debugging steps, and correlates failures with past known issues.

**AI-Driven Network Operations Assistant:** Designed and implemented AI-powered automation workflows leveraging domain-specific instructions, Copilot, Codex, and internal knowledge systems to provide regression insights, troubleshooting guidance, topology validation, infrastructure operations support, and automation recommendations.

**Cisco Regression Automation Framework:** Managed a regression automation ecosystem supporting 150+ devices and 500+ automation scripts across Cisco NXOS and IOS-XE platforms, including lab readiness, Jenkins job execution, Python environments, and continuous regression workflows.

## **KEY ACHIEVEMENTS**

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- Managed automation infrastructure with 150+ devices and 500+ regression automation scripts.
- Completed 300+ automation integration activities across Cisco enterprise networking platforms.
- Reduced regression failure triage time from 1-2 hours to approximately 15 minutes using AI-assisted analysis.
- Designed and deployed AI-powered regression support and failure analysis solutions used by engineering teams for troubleshooting and operational decision-making.
- Trained 10+ engineers on automation, regression workflows, AI-assisted troubleshooting, and Cisco platform debugging.
- Recognized with Certificate of Excellence for TCL to PyATS Migration, 4x Special Initiative Award, and Star of the Month Award.

## **EDUCATION AND CERTIFICATION**

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**Education:** Bachelor of Technology (B.Tech) in Information Technology - GB Pant University of Agriculture and Technology, 2011

**Certification:** AWS Certified Developer - Associate, 2022